The nature and variety of living organisms

Mark scheme and worked solutions for 10 questions: 3 multiple choice, 7 structured.

1 · Part 1

Paper 1 & 2 | Characteristics of living organisms | [1]

Answer: D

Worked solution

The eight shared characteristics are: nutrition, respiration, excretion, response to surroundings, movement, control of internal conditions, reproduction, and growth and development. Check each list and reject any list with an item that only some organisms show.

2 · Part 1

Paper 1 & 2 | Variety of living organisms | [1]

Answer: B

Explanation

Fungi cannot photosynthesise, so they have no chloroplasts; their cell walls are made of chitin and they may store carbohydrate as glycogen, so B is correct. A describes a plant cell. C describes an animal cell, which has no cell wall. D has a cellulose wall and starch store, which are plant features, not fungal features.

3 · Part 1

Paper 1 & 2 | Variety of living organisms | [1]

Answer: A

Explanation

Viruses are parasitic and can reproduce only inside living cells, so A is correct. B is wrong because viruses have no cellular structure: they have a protein coat, not a cell membrane and cytoplasm. C is wrong because a virus contains only one type of nucleic acid, either DNA or RNA. D is wrong because viruses are smaller than bacteria.

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4 · Part 1

Paper 1 & 2 | Characteristics of living organisms | [6]

Mark scheme

(a) **[4]**

- travels around the floor: movement (1)
- changes direction when it detects a wall: response to surroundings (1)

ALLOW sensitivity OR responds to stimuli

- takes in electrical energy at its base: nutrition (1)

ALLOW feeding

REJECT respiration

- fan switches on when the motor is too hot: controlling internal conditions (1)

ALLOW homeostasis OR temperature control OR keeping internal conditions constant

REJECT excretion

(b) **[2]**

- a living organism shows all of the characteristics of living organisms (1)
- the robot does not reproduce OR does not grow and develop OR does not respire OR does not excrete (1)

IGNORE "it is not made of cells" unless linked to a missing characteristic

Worked solution

(a) Match each behaviour to the closest characteristic. Travelling around the room is movement. Changing direction when a sensor detects a wall is responding to the surroundings. Taking in electrical energy to keep working is most like nutrition, because organisms take in substances to provide energy. Do not say respiration: respiration is the release of energy inside cells, not taking in an energy source. Switching on a fan when the motor is hot keeps conditions inside the robot steady, which is like controlling internal conditions.

(b) To be living, something must show all of the characteristics, not just some of them. The robot cannot reproduce (make new robots), does not grow and develop, does not respire and does not excrete waste, so it is not living.

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5 · Part 1 Paper 1 & 2 Variety of living organisms [7]

Mark scheme

(a) **[2]**

- bean seed contains starch (1)
- liver contains no starch (1)

ALLOW liver does not contain starch OR starch is absent from liver

REJECT liver contains no carbohydrate

(b) **[2]**

- plant cells contain chloroplasts but human cells do not (1)

ALLOW plant cells can photosynthesise but human cells cannot

- plant cells have cell walls made of cellulose but human cells have no cell walls (1)

ALLOW plant cells have a cell wall but human cells do not

ALLOW plant cells have a (large, permanent) vacuole but human cells do not

- each mark needs a comparison, or both sides clearly implied

(c) **[1]**

- they are able to move from one place to another (1)

ALLOW they can move around OR locomotion

ALLOW they (often) store carbohydrate as glycogen

(d) **[2]**

- iodine solution detects only starch (1)
- sucrose does not turn iodine solution blue-black OR the iodine solution would stay orange-brown even though carbohydrate is present (1)

ALLOW the root may contain little or no starch, so the test would be negative

Worked solution

(a) Blue-black shows starch, so the bean seed contains starch. The liver stayed orange-brown, so it has no starch. This does not mean liver has no carbohydrate at all: animals often store carbohydrate as glycogen, which iodine solution does not detect.

(b) Plant cells contain chloroplasts and can photosynthesise; animal cells cannot. Plant cells have cellulose cell walls; animal cells have no cell walls.

(c) Animals usually have nervous co-ordination and can move from one place to another; plants stay in one place.

(d) Iodine solution is a test for starch only. Sucrose is a different carbohydrate and does not give a blue-black colour, so the root would give a negative result even though it stores carbohydrate.

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6 · Part 1

Paper 1 & 2 | Variety of living organisms | [7]

Mark scheme

(a)(i) **[1]**

- hyphae (1)

ALLOW hypha

(a)(ii) **[1]**

- mycelium (1)

(b) **[3]**

- the *Mucor* secretes digestive enzymes out of its hyphae onto the agar (1)

ALLOW extracellular digestion OR enzymes released onto the food

- the enzymes digest OR break down the starch (1)
- the products of digestion are absorbed by the *Mucor* (1)
- so there is no starch left around the *Mucor*, so iodine solution does not turn blue-black there (1)

IGNORE the term saprotrophic nutrition on its own

- max 3

(c) **[2]**

- (30 − 6) ÷ (4 − 1) OR 24 ÷ 3 (1)
- 8 mm per day (1)

ALLOW 8.0 mm per day

- correct answer with no working scores 2

Worked solution

(a) Fungi such as *Mucor* are made of thread-like hyphae, which contain many nuclei. The network of hyphae is called a mycelium.

(b) Fungi feed by saprotrophic nutrition. The hyphae secrete digestive enzymes onto the food outside the fungus. In this agar the food is starch, so the enzymes break the starch down into smaller, soluble products, which the hyphae then absorb. Near the *Mucor* all the starch has been digested, so when iodine solution is added there is no starch to turn blue-black and a clear circle is seen. As the enzymes spread further, the circle gets bigger.

(c) Change in diameter = 30 − 6 = 24 mm. Time taken = 4 − 1 = 3 days. Rate = 24 ÷ 3 = 8 mm per day. Check: 6 + (8 × 3) = 30 mm, which matches day 4.

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7 · Part 1

Paper 1 & 2 | Variety of living organisms | [8]

Mark scheme

(a) **[2]**

- nucleus (1)

ALLOW mitochondria

- the DNA is in a circular chromosome (1)

ALLOW a loop of DNA OR circular DNA

IGNORE plasmids, as an extra

REJECT DNA in a nucleus

(b)(i) **[2]**

- the pH decreased from 6.6 to 4.5 (1)

ALLOW the pH decreased throughout, for the first mark only if no figures are given

- the decrease was fastest between 2 and 4 hours OR the decrease slowed down after 4 hours OR the pH levelled off at the end (1)

ALLOW the decrease was slow in the first hour

ALLOW fastest between 1 and 4 hours OR fastest between 2 and 3 hours

(b)(ii) **[2]**

- (5.9 − 4.8) ÷ (4 − 2) OR 1.1 ÷ 2 (1)
- 0.55 pH units per hour (1)

ALLOW 0.6 pH units per hour

- correct answer with no working scores 2

(c) **[2]**

- it is a control (1)
- to show that the decrease in pH was caused by the bacteria (1)

ALLOW to show that the pH does not fall without the bacteria OR to show that the change was not caused by the temperature or by the milk alone

- max 2

Worked solution

(a) Bacteria are prokaryotes: they have no nucleus. A yeast cell is a fungus cell, which does have a nucleus. In a bacterium the DNA is a single circular chromosome in the cytoplasm; bacteria may also have small rings of DNA called plasmids.

(b)(i) Give the overall trend with figures (6.6 down to 4.5), then describe how the rate changed: the pH fell slowly at first, fastest between 2 and 4 hours, and then only slowly after 4 hours as it levelled off.

(b)(ii) Change in pH = 5.9 − 4.8 = 1.1. Time taken = 4 − 2 = 2 hours. Rate = 1.1 ÷ 2 = 0.55 pH units per hour. Check: 4.8 + (0.55 × 2) = 5.9.

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(c) The second flask is a control. It is the same except that it has no bacteria. Its pH did not change, so the fall in pH in the first flask must have been caused by the bacteria producing lactic acid, not by the temperature or by the milk changing on its own.

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8 · Part 1

Paper 1 & 2 | Variety of living organisms | [8]

Mark scheme

(a) **[1]**

- an organism that causes disease (1)

ALLOW a micro-organism OR microbe that causes disease

ALLOW something (such as a virus) that causes disease

(b) **[4]**

- W: protoctist (1)
- X: bacterium (1)
- Y: virus (1)
- Z: fungus (1)

(c) **[3]**

- fewer chloroplasts so less photosynthesis (1)
- so less carbohydrate OR glucose OR sugar OR food is made (1)
- so less is available for energy OR for making new cells OR for growth (1)

ALLOW less starch can be stored, for the second mark

Worked solution

(a) A pathogen is an organism (or a virus) that causes disease. Pathogens may be fungi, bacteria, protoctists or viruses.

(b) W is single-celled with a nucleus but no cell wall, so it is a eukaryote but not a plant, animal or fungus: it is a protoctist (the malaria pathogen *Plasmodium* is like this). X has no nucleus, a circular chromosome and plasmids, which are features of bacteria. Y has no cellular structure, just a protein coat around one type of nucleic acid, so it is a virus. Z is made of hyphae with chitin walls, which are features of a fungus.

(c) Plants make their food by photosynthesis in chloroplasts. If chloroplasts do not form, the leaves carry out less photosynthesis, so the plant makes less carbohydrate. The plant then has less food to provide energy and to build new cells, so it grows less.

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9 · Part 1

Paper 1 & 2 | Variety of living organisms | [7]

Mark scheme

(a) **[2]**

- 24 ÷ 200 000 = 0.00012 mm OR 24 000 ÷ 200 000 (1)
- 0.12 µm (1)

ALLOW 1.2×10^{-1} µm

- correct answer with no working scores 2

(b) **[2]**

- 0.9 ÷ 0.12 (1)
- 7.5 (times) (1)

ECF from (a): 0.9 ÷ their answer to (a), correctly evaluated, for 2 marks

(c) **[3]**

- viruses have no cellular structure OR have no cytoplasm, cell membrane or nucleus (1)

ALLOW they are only a protein coat around nucleic acid

- they can reproduce only inside living cells OR cannot reproduce on their own (1)
- they do not show all the characteristics of living organisms (1)

ALLOW a named characteristic they do not show on their own, e.g. they do not feed OR respire OR excrete OR grow

- max 3

Worked solution

(a) Actual size = image size ÷ magnification = 24 mm ÷ 200 000 = 0.00012 mm. Convert to µm by multiplying by 1000: 0.00012 × 1000 = 0.12 µm. Check: 0.12 µm × 200 000 = 24 000 µm = 24 mm, the size in the image.

(b) 0.9 µm ÷ 0.12 µm = 7.5, so the bacterium is 7.5 times wider than the virus. This shows that viruses are smaller than bacteria. Both values must be in the same unit before dividing.

(c) Viruses are particles, not cells: they have a protein coat and one type of nucleic acid but no cytoplasm or membrane. They are parasitic and can only reproduce inside living cells. On their own they do not show the characteristics of living organisms such as nutrition, respiration, excretion and growth, so they are not considered to be living.

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10 · Part 1

Paper 1 & 2 | Variety of living organisms | [7]

Mark scheme

(a) **[5]**

Indicative points, 1 mark each, max 5:

- it has a nucleus, so it is not a bacterium (1)

ALLOW it is eukaryotic, so not a bacterium

- it is single-celled, but plants and animals are multicellular (1)
- it has chloroplasts OR it can photosynthesise, so it is not an animal (1)
- it has chloroplasts OR it can photosynthesise, so it is not a fungus (1)
- it has no cell wall, so it is not a plant, because plants have cellulose cell walls (1)
- it has no cell wall, so it is not a fungus, because fungi have chitin cell walls OR it is not made of hyphae (1)
- it has some plant-like features (chloroplasts) and some animal-like features (no cell wall, moves), like other protoctists (1)
- a point stating that a feature rules out a group gains credit only if the feature is correctly matched to that group

(b) **[2]**

- in the dark it cannot photosynthesise (1)
- so it can still obtain food OR energy OR nutrition (1)

ALLOW it does not depend on light for its food

Worked solution

(a) Work through each group. Bacteria have no nucleus, but *Euglena* has one, so it is not a bacterium. Plants and animals are multicellular, but *Euglena* is single-celled. *Euglena* has chloroplasts and photosynthesises, which animals and fungi cannot do. It has no cell wall, but plants have cellulose walls and fungi have chitin walls. So it fits none of these groups. Like other protoctists, it is a microscopic single-celled organism with a nucleus, and it has a mixture of plant-like features (chloroplasts) and animal-like features (no cell wall, able to move).

(b) Photosynthesis needs light. In the dark *Euglena* cannot make its own food, but because it can take in food particles it can still get the nutrients and energy it needs.

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